- The **Mid Term Exam 2** for **STA 2023** will be held in class on **November 3, 2026** (this mirror: time yourself, **50 minutes**).
- The exam will cover Chapters 4-6.
- The exam is **closed-book and closed-notes**.
- You may use a **calculator** and **one double-page cheat sheet**.
- **You must bring your physical UCF ID card. Your exam will not be graded if you fail to present your physical UCF ID card.**
- You are **not allowed** to use laptops, cellphones, tablets, or any other electronic devices.
- All cellphones must be **turned off** before the exam begins.
- Show all necessary steps for full credit.
- **Write all work clearly.** Answers without supporting steps may not receive credit and Round all probabilities to **four decimal places**.
- **Write answers in the spaces provided.** If you need additional space, clearly label the continuation.
- All needed z-scores and standard normal areas are given as **hints** on the exam paper.
- **Total points: 50.**

Question 0: Academic Integrity Acknowledgment (Required)

As your first task, please **handwrite the acknowledgment statement below** on your answer sheet. Then write your **full name**, **signature**, and **UCF ID number (in digits)** below it.

Acknowledgment Statement:

> "I acknowledge that I have read and understood all exam instructions and that I have fully complied, and will continue to comply, with the university's policies on Academic Integrity in completing and submitting this exam."

Name (print clearly): _____

Signature: _____

UCF ID Number: _____

Question 1 (6 points)

Commuting and Housing

The table below summarizes how 110 students at a university commute and where they live.

Commute	Lives on campus	Lives off campus
Drive	8	34
Bus	10	16
Walk	22	4
Bike	9	7

Total on campus = 49, Total off campus = 61, Overall Total = 110

Answer each question. Show your work when appropriate.

(a) **(1 points)** What is the probability that a randomly selected student lives off campus?

(b) **(1 points)** What is the probability that a randomly selected student walks to campus?

(c) **(1 point)** What is the probability that a randomly selected student drives OR lives on campus?

(d) **(1 points)** What is the probability that a randomly selected student bikes AND lives off campus?

(e) **(2 points)** What is the probability that a randomly selected student takes the bus given that the student lives off campus?

Question 2 (2 points): Discrete Probability Distribution

Determine whether the following distribution represents a valid **discrete probability distribution**. If not, explain why.

x	$P(x)$
1	0.15
2	0.25
3	0.30
4	0.20
5	0.12

Question 3 (8 points): Cars per Household

A city planner records the number of cars, X, owned by each household in a random sample of 85 households in one neighborhood. The following data represent the number of cars per household. *Note: no household in the sample owned more than 3 cars.*

x	Frequency
0	13
1	29
2	33
3	10

(a) (2 pt) Construct a probability distribution for the random variable X, the number of cars per household.

(b) (2 pt) Graph the discrete probability distribution.

(c) (2 pt) Compute and interpret the mean of X.

(d) (2 pt) Compute the standard deviation of X.

Question 4 (8 points): Reading Habits

According to a national survey, 35% of U.S. adults read at least one book last month. In a random sample of 8 U.S. adults, find the probability that:

(a) (2 pt) Exactly 2 read at least one book last month.

(b) (2 pt) At most 2 read at least one book last month.

(c) (2 pt) At least 1 read at least one book last month.

(d) (2 pt) For a random sample of 300 U.S. adults, find the expected number and standard deviation of those who read at least one book last month.

Question 5 (5 points): Shuttle Wait Time

Suppose the waiting time X (in minutes) for a campus shuttle is uniformly distributed between 0 and 25 minutes.

(a) (1 pt) Sketch or describe the graph of the uniform probability density function (pdf).

(b) (2 pt) Find the probability that the wait is between 5 and 15 minutes, i.e., $P(5 \le X \le 15)$.

(c) (2 pt) Find the probability that the wait is more than 18 minutes, i.e., $P(X > 18)$.

Question 6 (6 points): Normal Distribution and z-Scores

Instructions: Use the hints below (or the standard normal table) to find each probability. Show work and round final answers to four decimal places.

Hints: $\mathbf{P(Z < -0.85) = 0.1977}$, $\mathbf{P(Z < -1.30) = 0.0968}$, $\mathbf{P(Z < -0.25) = 0.4013}$, $\mathbf{P(Z < 1.25) = 0.8944}$.

(a) Find the area to the left of $z = -0.85$.

(b) Find the area between $z = -1.30$ and $z = -0.25$.

(c) The random variable X is normally distributed with $\mu = 250$ and $\sigma = 40$. Find $P(X > 300)$.

Question 7 (8 points): Heights of Adult Men

The heights of adult men in the U.S. are approximately normally distributed with mean $\mu = 69$ inches and standard deviation $\sigma = 2.8$ inches.

Hints: $\mathbf{P(Z < 1.79) = 0.9633}$, $\mathbf{P(Z < -1.43) = 0.0764}$, $\mathbf{P(Z < -1.07) = 0.1423}$, $\mathbf{P(Z < 1.07) = 0.8577}$; the 10th percentile corresponds to $\mathbf{z = -1.28}$.

(a) (2 pt) What is the probability that a randomly selected man is taller than 74 inches?

(b) (2 pt) What is the probability that a randomly selected man is shorter than 65 inches?

(c) (2 pt) What proportion of men are between 66 and 72 inches tall?

(d) (2 pt) What height places a man at the 10th percentile?